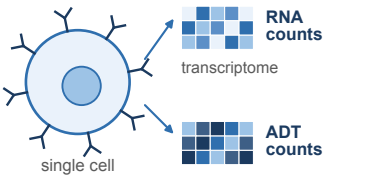
A Paired CITE-seq input



PBMC10k | PBMC5k

Paired RNA and protein measured in the same cells

B Representation framework

RNA only

- **RNA PCA** — RNA-only linear reference — *linear / PCA*
- **scVI_matched** — capacity-matched RNA-only VAE — *deep generative*

RNA + protein (multimodal)

- **Concat RNA–protein PCA** — simple multimodal feature-access reference — *linear / PCA*
- **totalVI** — existing multimodal model; Gayoso et al. — *deep generative*

Common downstream classifier: logistic regression on each representation → cell-type labels

C Comparison logic

Descriptive contrasts; not causal architecture effects.

RNA PCA (RNA-only reference) → **Concat RNA–protein PCA** (simple multimodal reference) → **totalVI** (complex multimodal representation)

Protein-access gain

$$G_{access} = F1_{concat} - F1_{RNA\ PCA}$$

Residual totalVI-associated contrast

$$G_{assoc} = F1_{totalVI} - F1_{concat}$$

Central question: how much of the observed multimodal gain is associated with protein access, and does a residual totalVI-associated contrast remain beyond a simple RNA–protein representation?

D Internal evaluation protocol

9,494 high-confidence PBMC cells (development cohort)

Training cells (n = 7,573) | **Held-out (n = 1,921)**

1. All unsupervised fits (HVG, scaling, PCA, scVI_matched, totalVI) use training cells only
2. Held-out cells are encoded after fitting
3. Logistic regression is trained on training labels
4. Macro-F1 is evaluated on held-out cells

Strict train-only information access (primary)

E Bidirectional transfer

Direction A	PBMC10k source	PBMC5k target
Direction B	PBMC5k source	PBMC10k target

- **Inductive concat PCA** — source-only fit; target transformed unchanged
- **Feature-access-matched transductive concat** — unsupervised fit on source + unlabeled target features
- **scVI_matched + scArches** — unsupervised query adaptation
- **totalVI + scArches** — unsupervised query adaptation

Target labels are never used during adaptation.

Feature-access-matched concat matches unlabeled target-feature access, not the scArches learning objective or neural adaptation algorithm.

F Reliability and robustness analyses

Protein corruption
training-time (retrain) and post-adaptation (target only)

Predictive uncertainty
$U_{pred,i} = 1 - \max_k p(y_i = k \mid z_i)$
from the downstream classifier

Latent posterior dispersion
representation-level dispersion; not formal epistemic uncertainty

Risk–coverage curves
selective prediction: defer uncertain cells; track retention

Training stochasticity

source subsets × model seeds × run replicates → variance decomposition

G External donor validation

Lawlor blood CITE-seq: 10 biological donors

- Leave-two-donors-out folds; each donor held out exactly once
- Aggregation unit: biological donor (n = 10)
- 5 representations: RNA PCA, protein PCA, concat PCA, scVI_matched, totalVI

Lawlor author labels are protein-gated.

Outputs Macro-F1 · G_{access} / G_{assoc} with paired-bootstrap intervals · transfer contrasts · degradation trajectories · risk–coverage · donor-level differences